

# Global Weather Pattern Analysis and Prediction Using the Elastic Stack and Machine Learning

Authors: Andrea Garcia-Tellez, Felix Garcia, Karla Salazar, Kyle Giang,  
Department of Information Systems, California State University, Los Angeles

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[agarc666@calstatela.edu](mailto:agarc666@calstatela.edu), [fgarci172@calstatela.edu](mailto:fgarci172@calstatela.edu), [ksalaz63@calstatela.edu](mailto:ksalaz63@calstatela.edu), [tgiang12@calstatela.edu](mailto:tgiang12@calstatela.edu)

**Abstract:** This project will use the Global Weather Repository dataset from Kaggle (which was converted into a [JSON file](#) [1]). This dataset provides daily weather information for major cities and has a size of approximately 25.93MB. We'll begin by ingesting this data into an Elasticsearch cluster to enable efficient indexing and querying. This will help us to conduct rapid and large-scale data exploration, allowing us to uncover key insights. For example, trends in temperature, weather conditions, humidity, regional differences in air quality, and the relationships between various weather metrics. The second phase of the project will involve applying machine learning to this dataset. We'll train a predictive model to forecast a key weather variable, such as maximum daily temperatures, for specific cities. Finally, we'll present our findings, demonstrating both the power of Elasticsearch for comprehensive data analysis and the effectiveness of machine learning in making accurate weather predictions.

## 1. Introduction

The real-time analysis of global weather data has become crucial for enabling proactive decision-making across various key sectors, particularly in relation to human health and infrastructure durability. Traditional weather analysis often relies on fragmented or delayed data sources, which are inadequate for generating dynamic, granular insights required to take effective action, especially in the rapidly evolving global climate conditions. To address this issue, we need a unified and robust system that can efficiently manage vast amounts of data and perform complex analyses.

### 1.1 Background and Importance

The project's significance stems from the concerns about global change, as reflected in the climate data. The purpose is to anticipate and mitigate global vulnerabilities. Understanding the complex relationships between features such as temperature, humidity, UV index, and air quality is crucial for predictive risk modeling. The utilization of the Elastic Stack for this project serves as a vital case study in big data analytics, demonstrating the efficacy and scalability of modern, large-scale data processing tools in a high-stakes domain.

### 1.2 Project Methodology and Objectives

This project addresses the challenges of conventional approaches by deploying a resilient and scalable data ingestion system, as well as a visualization pipeline using the Elastic Stack (Elasticsearch and Kibana). We leveraged

the Global Weather Repository dataset from Kaggle, a collection of daily weather metrics for capital cities worldwide, to establish a high-performance analytical foundation for this investigation.

This project is structured into two connected phases, as outlined in the Abstract. We designed it to transform this raw data into actionable, geographic, and comparative insights:

1. **Exploratory Data Analysis (EDA):** The first phase would focus on visualizing global weather patterns to uncover critical regional trends. This would involve the identification of:
  - a. Regions and cities are experiencing the highest cumulative heat and humidity levels.
  - b. Geographic distribution of intense UV exposure and associated sun-related health risks.
  - c. Global areas exhibiting severe air quality degradation, as measured by the standardized US EPA index.
2. **Machine Learning (ML) Prediction:** The second phase will utilize Kibana's integrated machine learning capabilities to develop and train a predictive model. This model would be able to predict a critical factor, such as the maximum daily temperature. This shows that we can use ML on massive datasets to give us precise, future-focused estimates of risks.

By integrating the scalable indexing capabilities of Elasticsearch with the visualization and ML power of Kibana, this work successfully validates how modern data infrastructure can provide both immediate, dynamic analysis and powerful predictive modeling.

## 2. Related Work

Using weather data in various forms is not a new concept. In fact, using weather data for forecasting is also well established. What differs is how the data is used to make predictions, as different methods and tools can produce different results. This project uses the Global Weather Repository dataset from Kaggle, which is uploaded into Elasticsearch with Kibana to create visualizations and perform machine learning analysis. The following works use other tools and methods to forecast weather, and

discussing these approaches will show how this project differs from them.

One study uses Python and artificial neural networks to forecast short-term temperatures by pulling data from online APIs. The goal is to train ANN models that predict temperature several hours ahead and test how additional features, such as precipitation, affect accuracy. This approach depends on custom Python scripts and external ML code. In contrast, this project stores the weather dataset directly inside Elasticsearch, indexes it for fast search, and uses built-in ML tools instead of separate Python models.[2]

Another paper uses Hadoop and MapReduce to process decades of historical weather data from many U.S. cities and identify "proper weather days" for event planning. Their work focuses on large-scale batch processing through distributed computing. This method differs from the project in that Hadoop runs offline batch jobs. In contrast, Elasticsearch supports real-time indexing and immediate visualization in Kibana, including maps and charts built directly on the indexed data.[3]

A third study evaluates machine learning forecasting models, such as ERA5 and PanguWeather, and compares their performance to that of traditional numerical weather prediction systems. The researchers analyze long-range weather behavior and how ML handles global patterns and extreme events. This differs from this project because the ML work in the paper relies on external forecasting models. In contrast, the project performs anomaly detection and regression inside Elastic Cloud using the ingested dataset.[4]

While the related works use weather data for forecasting, none of them leverage Elasticsearch or Kibana for integrated data management and analysis. Their pipelines rely on Python scripts, Hadoop batch jobs, or external ML frameworks, rather than a real-time indexing engine or interactive dashboards. In contrast, this project uploads the Global Weather Repository dataset from Kaggle into Elasticsearch, defines field mappings, and indexes the data for efficient bulk ingestion. A Kibana Data View enables interactive visualizations, such as charts, maps, and dashboards, directly on the indexed data. The dataset is also uploaded to Elastic Cloud for machine learning, supporting multi-metric anomaly detection and regression models for temperature prediction. Overall, this project demonstrates a fully end-to-end workflow, from data ingestion and cleaning to visualization and integrated ML, all within a single platform, showing a clear distinction from the approaches used in previous studies.

### 3. Specifications

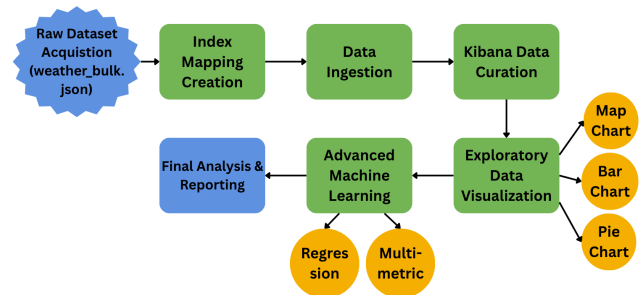
The platform is built on the Elastic Stack ecosystem, utilizing Elasticsearch as the core data engine and Kibana for visualization and machine learning workflows. The Elasticsearch and Kibana versions used were 8.9.1, with the LocalHost deployment running on either Windows or Mac. The current environment is deployed as a single-node cluster to manage the end-to-end data pipeline. **Table 1** summarizes the technical specifications of the cluster and the underlying host machine:

*Table 1- Cluster Specifications*

| Specification               | Value |
|-----------------------------|-------|
| Cluster Version             | 8.9.1 |
| Number of Nodes             | 1     |
| Total System Memory (RAM)   | 16 GB |
| Available Processors (Core) | 16    |

### 4. Implementation Flowchart

The implementation process is followed by a structured workflow, starting with downloading the Global Weather Repository dataset from Kaggle [5]. Elasticsearch and Kibana were initialized to form an analytical environment, where an index in Elasticsearch and data mappings were made to ensure accurate numeric and geospatial fields. The processed dataset was then uploaded to Elasticsearch for indexing and querying using GitBash. The Exploratory Data Analysis (EDA) took place in Kibana. This involved creating data views, verifying, and adjusting geolocation fields to support analysis and map-based visuals. As a result, numerous Kibana visualizations and dashboards were made to explore topics such as weather trends and regional patterns. Finally, the same structured process was extended for Elastic Cloud Deployment and Machine Learning. The weather\_bulk.json file was uploaded to Elastic Cloud using Gitbash, which created the foundation for developing and training machine learning algorithms and predictive models.



*Figure 1- Implementation Flowchart*

### 5. Analysis and Visualizations

Following the data refinement process, the Global Weather dataset was utilized to conduct a comprehensive, multi-faceted analysis of global climatic patterns. This section serves two primary functions: first, to present the core findings through data visualizations via Kibana, mapping things like spatial and temporal distribution of heat, humidity, and atmospheric quality; and second, to employ Machine Learning (ML) models within Elastic Cloud to diagnose anomalies and develop a predictive regression model for temperature.

#### 5.1 Exploratory Data Visualization (Kibana)

### 5.1.1 Global Conditions and Thermal Extremes

The initial visualization provided a summary of dominant weather types and concentrated heat areas. A Pie Chart analysis of global weather conditions revealed that Partly Cloudy (36.22%), Sunny (33.23%), and Patchy rain nearby (9.18%) were the three most frequent conditions, with severe weather being less typical worldwide during the six months (May 10 – Oct 31, 2024), indicating a dominance of clear or semi-clear weather.

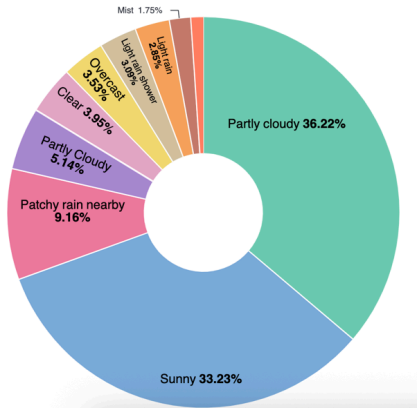


Figure 2 - Global Distribution of Weather Conditions (May 10 - Oct 31, 2024)

Correspondingly, a second Pie Chart focusing on cumulative heat, measured by the sum of *feels\_like\_celsius*, isolates extreme thermal environments. This analysis (Figure 2) identified a clear cluster of high-sensory temperature cities, primarily in arid environments, with Djibouti City (11.11%), Muscat (10.93%), Islamabad (10.16%), and Kuwait City (10.09%) ranking highest for total cumulative heat experienced. The analysis confirms that the cities with the most cumulative heat are heavily concentrated in the Middle East and South Asia.

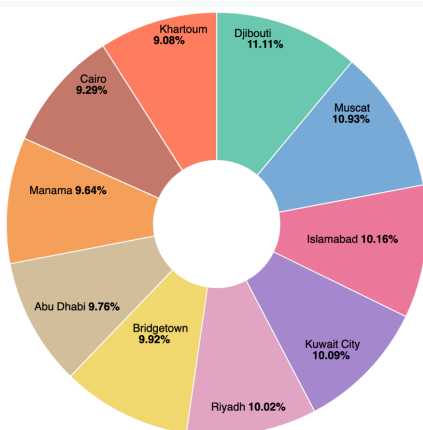


Figure 3 - Top 10 Cities Experiencing the Most Cumulative Heat (May 10 - Oct 31, 2024)

### 5.1.2 Regional Extremes in Humidity and UV Exposure

Bar charts were made to identify localized environmental extremes. The Vertical Bar Chart detailing the *Top 10 Cities with the Highest Average Humidity* highlighted coastal and equatorial locations, including Meyungs, San

Salvador, New Guatemala. Yopal, Georgetown, Santo Domingo, Douala, Paramaribo, Havana, and 'S Gravenstaftei. This demonstrates that tropical and coastal regions, specifically Central America, the Caribbean, and Equatorial Africa, are characterized by higher average humidity. As a result of their proximity to warm ocean currents and high rates of evaporation, they have a higher average atmospheric moisture content.

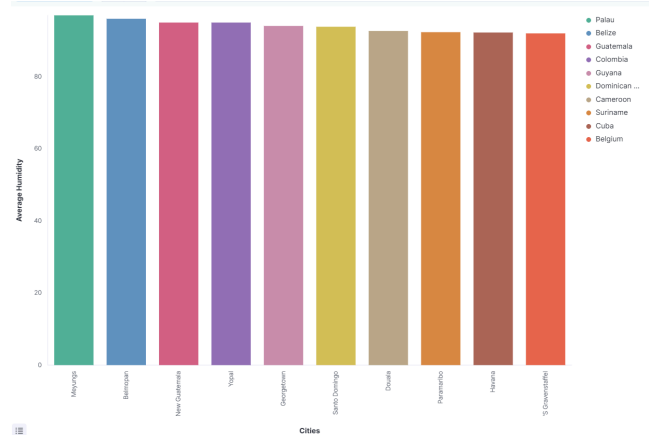


Figure 4 - Top 10 Cities with the Highest Humidity (May 10 - Oct 31, 2024)

In contrast, the analysis of solar intensity, via a Vertical Bar Chart showing the Top 10 Countries with the Highest Average UV Index (Figure 4), showcased countries in the Middle East, Central Asia, and Equatorial Africa such as Saudi Arabia, Iraq, Kuwait, Turkmenistan, Democratic Republic of Congo, Republic of Congo, Qatar, Tanzania, Sudan, and Djibouti as having the most intense solar exposure. This finding demonstrates that intense solar exposure is heavily concentrated across the world's arid, low-latitude regions, which provide minimal natural protection against solar intensity.

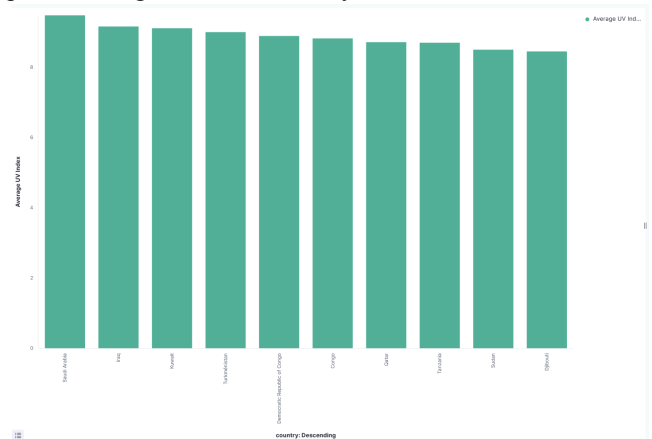
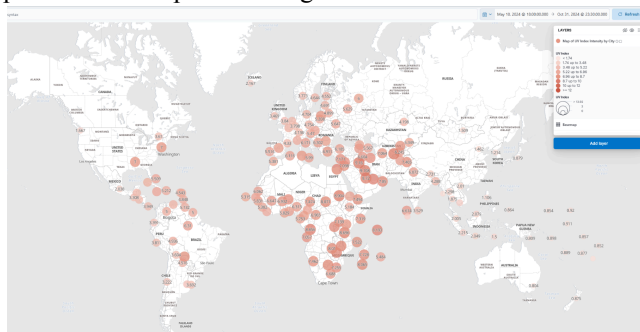


Figure 5 - Top 10 Countries with the Highest Average UV Index (May 10 - Oct 31, 2024)

### 5.1.3 Spatial Analysis of Atmospheric Conditions

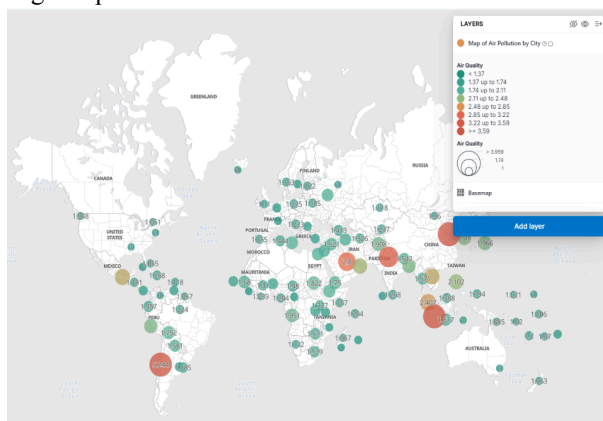
The analysis was further enhanced by geographical visualizations, which focused on tempo-spatial (time and geo-location) information. Two Kibana Maps provided a spatial understanding of atmospheric intensity. *The Map of UV Index Intensity by City* (Figure 5) utilized a red color

gradient and cluster size to indicate intensity, making high-risk areas immediately visible. This visualization demonstrates that the highest UV intensity (indicated by the darkest and largest clusters) is concentrated in low-latitude regions, particularly across Saudi Arabia and Central Africa. This finding suggests that the intense solar exposure is a direct reflection of the dry and sunny conditions in these regions, which, in combination with their minimal cloud cover and atmospheric moisture, provide limited protection against solar radiation.



**Figure 6 - The Map of UV Index Intensity by City (May 10 - Oct 31, 2024)**

Similarly, the *Map of the Air Pollution by City* (Figure 6) employed a green-to-red gradient (adhering to US EPA standards) to spatially identify specific urban and industrial hotspots that exceeded healthy air quality thresholds, providing an essential view of environmental air quality worldwide. This visualization shows that, although overall global air quality remains generally good, specific regions, such as China, Chile, Pakistan, Saudi Arabia, and Indonesia, experience significantly poorer air quality. This pattern identifies areas where high population density, industrial activity, or specific geographical features (e.g., mountain ranges that trap pollutants) contribute to increased atmospheric contamination, necessitating targeted public health interventions.

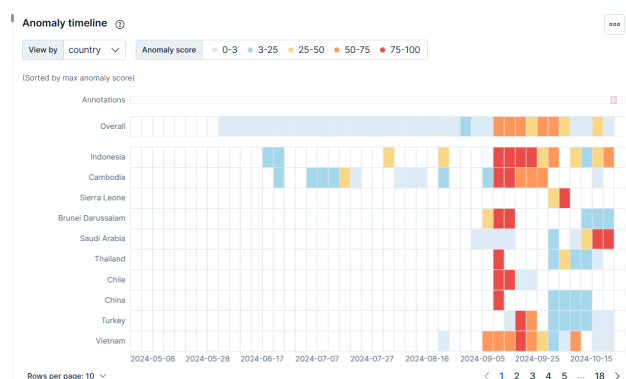


**Figure 7 - The Map of the Air Pollution by City (May 10 - Oct 31, 2024)**

## 5.2 Machine Learning and Predictive Analysis (Elastic Cloud)

### 5.2.1 Multi-metric Anomaly Detection (Temporal)

To identify deviations from expected weather patterns, a Multi-metric Anomaly Detection Job (Figure 8) named *multimetroc\_location\_outlier\_detector* was created. This analysis is temporal, as it allows for the observation of anomalies to be viewed by specific dates (e.g., 2024-05-08, 2024-05-28), as shown in the timeline. The model continuously monitored the mean values for temperature, humidity, UV index, and air quality on a daily bucket span (1d), splitting the analysis by city (location\_name). The daily bucket span enables this model to function as a time-series job, where the historical norm is calculated and compared against new data on a daily basis. As shown in the results (Figure 7), the model's primary function is to serve as a diagnostic tool, automatically flagging instances where a city's environmental profile deviates significantly from its historical norm, which could signal an emerging extreme weather event or a data anomaly. The analysis of the detected anomalies further revealed the top influencers (fields) most frequently associated with an extreme deviation, which were country, humidity, and UV index—suggesting that localized weather-related factors and the broader geographical context are the most critical predictors of an unexpected change in a city's weather profile.



**Figure 8 - *multimetroc\_location\_outlier\_detector* (May 10 - Oct 31, 2024)**

### 5.2.2 Regression Model- Temperature Celsius

A Regression Model (Figure 9) was deployed to quantify the relationship between various environmental factors and the resulting temperature. This supervised learning job targeted *temperature\_celsius* as the dependent variable, while excluding redundant and data-leaking variables. The model featured a total feature importance section that indicates the degree to which each field affects the dependent variable, with the top four fields being latitude, UV index, wind speed (in kph), and precipitation (in mm). The model's accuracy was assessed against a testing dataset, yielding a high Coefficient of Determination (R-squared) of 0.927 and a Mean Squared Error (MSE) of 5.9. The resulting model (Figure 8) confirms that nearly 93% of the variance in temperature is explained by the selected feature set and that predictions are off by 2.45°C. This established a robust and highly accurate quantitative model for predicting global temperature within the dataset's boundaries.



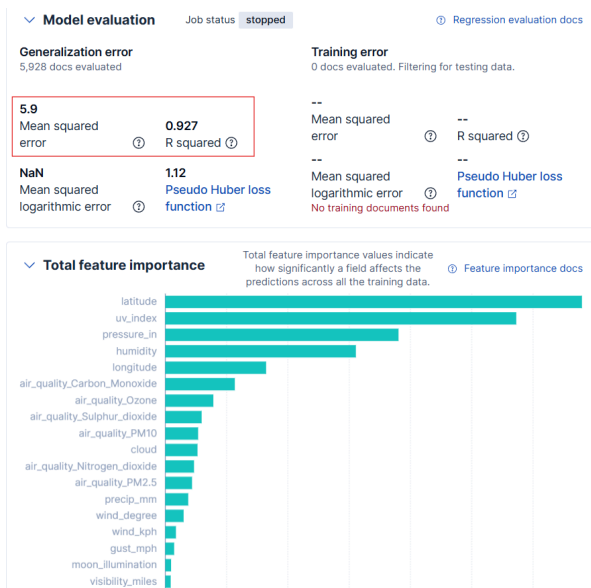


Figure 9 - Temperature\_Celsius\_Regression (May 10 - Oct 31, 2024)

## 6. Conclusion

This project has demonstrated how Elasticsearch [6] and Kibana can be utilized for large-scale analysis of weather data and predictive modeling. By ingesting the Global Weather Repository data into Elasticsearch, we have created a scalable system of processing and visualizing global climate trends. With Kibana, we have seen the patterns in humidity, air quality, temperature, UV intensity, and environmental risks. The machine learning aspect has further enhanced the analysis by creating and deploying a regression model to predict temperature values, achieving an R-squared of 0.927 and a mean squared error of 5.9. In addition to predictive modeling, the project successfully deployed a Multi-metric Anomaly Detection model to continuously monitor and flag significant deviations in the mean values for temperature, humidity, UV index, and air quality. This temporal diagnostic tool provided real-time alerts for emerging extreme weather events or data anomalies.

This project has demonstrated the power of Elasticsearch and Kibana by processing complex environmental data and transforming raw records into valuable insights and accurate forecasts. This is significant because it combines big data analytics with ecological knowledge, demonstrating how cloud-based tools like Elastic Cloud can support and monitor climate predictions in real-time.

## References

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<sup>1</sup> GitHub Link (for downloadable JSON file): <https://github.com/ksalaz63/GlobalWeatherAnalysis>